

```
25. apple['Stochastic_K'] = slowk
26. apple['Stochastic_D'] = slowd
27.
```

The advantages of using TA-Lib include:

- **Performance:** Much faster than custom Python implementations
- **Accuracy:** Well-tested implementations of standard indicators
- **Completeness:** Includes a vast array of indicators and candlestick pattern recognition

While installing TA-Lib can be challenging on some platforms, the performance benefits make it worth the effort for serious algorithmic traders.

*?? **Pro Tip:** While libraries like TA-Lib make calculating indicators easy, professional quants at firms like Two Sigma and AQR Capital often customize their technical indicators to gain an edge. For example, instead of using standard 14-day RSI, they might optimize the period for specific assets or combine the RSI with other factors. As you progress in your algorithmic trading journey, consider experimenting with modifications to standard indicators—adjusting periods, combining indicators in novel ways, or even creating entirely new indicators based on your market observations. This "indicator innovation" is one way quantitative traders continue finding edges in highly competitive markets.*

Using pandas-ta: A Pythonic Alternative

If you find TA-Lib difficult to install, pandas-ta is an excellent alternative that integrates seamlessly with pandas DataFrames:

```
1. # Installation (you only need to do this once)
2. # pip install pandas-ta
3.
4. import pandas_ta as ta
```